Sahaj Bhadja

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Summary

Full-stack Software Engineer with 2+ years shipping production **AI products** end to end, from **React** and TypeScript UIs to Python and Node.js APIs and multi-step **LLM workflows**, plus internal tools that remove manual work.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Advanced NLP, Database Systems, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Object-Oriented Programming, Operating Systems, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Built fault-tolerant **LLM orchestration** with FastAPI background workers, explicit job states, and failure isolation for an AI education product, processing 1K+ AI jobs at 3x throughput under concurrent load.
- Owned the multi-tenant data layer on Azure SQL with row-level security for 500+ users, cutting dashboard latency from 1.2s to 350ms, and automated **integration pipelines** running 5K+ monthly jobs at 99%+ reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Shipped an AI grading product **end to end**, from **React** UI and Flask REST APIs to MySQL persistence and AWS deployment, serving 150+ instructors and 10K+ daily requests in production.
- Built multi-step **LLM workflows** that generate rubrics, read handwritten answers via vision OCR, and grade with rationale across GPT-4o, Claude, and Qwen-VL, cutting feedback turnaround 60% at 90%+ consistency.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Reduced **manual workflows** by building cross-platform data-sharing microservices with automated flows, cutting manual data entry 35% and improving API response times 30% for low-bandwidth users through pagination and caching.
- Automated ETL on AWS Glue and S3 plus nightly Jenkins builds running 20+ PyTest and Mockito suites, boosting pipeline efficiency 68% while **collaborating** with Microsoft and GitHub stakeholders in Agile sprints.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built **internal tools** in React, TypeScript, and Node.js for lead management, billing, customer records, and resource allocation, with real-time WebSocket updates syncing 2M+ records across multiple locations.
- Used Google Analytics on feature usage and drop-off points to prioritize product fixes that cut resource wastage 42% and lifted **lead conversions** 15% through Jenkins CI/CD releases.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Designed the product architecture and **presented it to the client**, then built Flask booking, pricing, and coupon APIs plus a React TypeScript dashboard with sub-200ms responses across 50K+ records.
- Automated refunds, disputes, and payment-status updates through idempotent webhooks, cutting **manual** refund handling about 70%, and built hourly Spark analytics on revenue and active users for internal dashboards.

Skills

Languages: TypeScript, JavaScript, Python, SQL, Java, HTML, CSS
Frontend: React, Next.js, TailwindCSS, Shadcn, React Native, WebSockets
Backend: Node.js, Express.js, FastAPI, Flask, REST APIs, Microservices, Convex, Clerk
AI: LLM Systems, GPT-4o, Claude, Prompt Engineering, OCR, FAISS, Pinecone, Semantic Search
Data & Infra: PostgreSQL, MySQL, MongoDB, Firebase, AWS, Azure, Docker, GitHub Actions, CI/CD, Stripe, Git

Projects

AI Interview Synthesis Agent | *TypeScript, Next.js, Python, FastAPI, LLM APIs, PostgreSQL*

- Designed an **AI agent** that drafts interview guides from research goals, transcribes video answers, tags themes with verbatim quotes, and writes evidence-linked summaries, with an eval set catching **hallucinated quotes**.

Ops Automation Agent | *TypeScript, Next.js, Node.js, LLM Tool Calling, PostgreSQL, Stripe*

- Designed an **internal tool** where an LLM agent reconciles Stripe invoices against CRM deals, flags mismatches for finance, and drafts support replies, with human approval before any action and full audit history.

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, Clerk, Convex, Shadcn*

- Built an **AI SaaS** product for secure document upload, semantic search, summarization, and real-time collaborative document conversations with Clerk, Convex, and Shadcn UI, cutting document review time 60% for 200+ users.